2.11 Math module

Settings

Beta features

math module

Python comes with a standard math module to support such adva

module

A module is Python code located in another file.

Figure 2.11.1: Importing the math module and calling a math module functior

```
import math

num = 49
num_sqrt = math.sqrt(num)
```

function

A function is a list of statements that can be executed simply by re name.

function call

The process of invoking a function is referred to as a function call.

argument

The item passed to a function is referred to as an argument.

Try 2.11.1: Example of using a math function: Savings interest program.

Note: Blank print statements are used to go to the next line after reading pre-entered input.

Run

```
import math

base = float(input("Enter initial savings: "))
print()

rate = float(input("Enter annual interest % rate: "))
print()

years = int(input("Enter years that pass: "))
print()

total = base * math.pow(1 + (rate / 100), years)

print(f"Savings after {years} years is ${total:.2f}")
```

CONSOLE

Table 2.11.1: Functions in the standard math module.

Function	Description		Function	
Number representation and theoretic functions				
ceil(x)	Round-up value		fabs(x)	Absolute v
factorial(x)	factorial (3! = 3 * 2 * 1)		floor(x)	Round-do
fmod(x, y)	Remainder of division		fsum(x)	Floating-p
Power, exponential, and logarithmic functions				
exp(x)	Exponential function e^x		log(x, (base))	Natural log
pow(x, y)	Raise x to power y		sqrt(x)	Square roo
Trigonometric functions				
acos(x)	Arc cosine		asin(x)	Arc sine
atan(x)	Arc tangent		atan2(y, x)	Arc tanger
cos(x)	Cosine		sin(x)	Sine
hypot(x1, x2, x3, ..., xn)	Length of vector from origin		degrees(x)	Convert fr
radians(x)	Convert degrees to radians		tan(x)	Tangent
cosh(x)	Hyperbolic cosine		sinh(x)	Hyperbolic
Complex number functions				
gamma(x)	Gamma function		erf(x)	Error funct
Mathematical constants				
pi (constant)	Mathematical constant 3.141592...		e (constant)	Mathemat

CHALLENGE ACTIVITY 2.11.1: Math functions.

763650.1443318.qx3zqy7

Start

Type the program's output

```
import math

x = math.sqrt(4.0)
print(x)
```

2.0

1 2 3 4 5

Check Next

View solution *(Instructors only)*

CHALLENGE ACTIVITY 2.11.2: Using math functions to calculate the distance between two points.

763650.1443318.qx3zqy7

Organize the correct code statements to find the distance between point (x1, y1) and point (x2, points_distance. The calculation is:

$$Distance = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

Click here for example

Note: Not all code statements on the left should be used in the final solution.

How to use this tool

Unused

```
x_dist = math.pow((x1 - y1), 2)
x_dist = math.pow((x2 - x1), 2)
y_dist = math.pow((y2 - y1), 2)
y_dist = math.pow((x2 - y2), 2)
points_distance = math.sqrt(x_dist + y_dist)
```

main.py

```
import math

x1 = float(input())
y1 = float(input())
x2 = float(input())
y2 = float(input())

print(points_distance)
```

Check

View solution *(Instructors only)*

CHALLENGE ACTIVITY 2.11.3: Writing math calculations.

763650.1443318.qx3zqy7

Start

Compute: $answer = |b| - c$

Click here for example

Note: Functions in the math module can be accessed using dot notation. Ex: The constant pi is

```
1 import math
2
3 b = float(input())
4 c = float(input())
5
6 """ Your code goes here """
7
8 print(f"answer = {answer:.1f}")  # Outputs answer with 1 decimal place
```

1 2 3

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